

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC401 - BUILDING PLANNING AND DRAWING

Time : 1.5 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1	Which principal guides for position of door & window on external wall?
	a. Aspect
	b. Prospect
	c. Form
	d. Climate
	_____ Is the overall experience or feel of the room?
	a. Aspect
	b. Prospect
	c. Roominess
	d. Climate
3	Which rating system is used for green building in India?
	a. BREEAM
	b. LEED®
	c. GRIHA
	d. CASBEE
	Comparative study of the human body's measurements and capabilities is called _____?
	a. Physiology
	b. Anthropometry
	c. Biology
	d. Zoology
5	Which trap is provided at the junction of house drain and street sewer?
	a. Nahani
	b. Gully trap
	c. Intercepting
	d. Q trap

6	025336255	Which method is used for lighting design in building?	
		a. Lighting method	
		b. Lumen method	
		c. Lux method	
7	025336255	What is the recommended value air change / hr. for living room?	
		a. 3-6	
		b. 8-12	
		c. 2-4	
8	025336255	Which part is used in air conditioning for dust control?	
		a. exhaust fan	
		b. compressor	
		c. damper	
9	025336255	Travel distance between two staircases in residential building should be ____?	
		a. 22.5	
		b. 30 m	
		c. 25m	
10	025336255	Reverberation time for Auditorium should be	
		a. 20 s	
		b. 1 s	
		c. 2 s	
11	025336255	Arrangement of functional spaces as per priority and interdependency is called ____?	
		a. Aspect	
		b. Prospect	
		c. Grouping	
12	025336255	Horizontal angle of sun along the horizon is called ____?	
		a. Azimuth	
		b. Altitude	
		c. acute angled.	
13	025336255		

Q2. Solve the following.

A) Describe any three principals of Planning?

6

B) What are the bye laws for size, area & height of habitable rooms?

6

Q3. Solve the following.

- A) Which documents are required to take building permission from local authorities ? 6
- B) Define anthropometry with its importance in planning? 6
- Q4. Solve Any Two of the following.
- A) Describe Nahani trap & gully trap with sketch? 6
- B) Explain Lumen method of lighting design with an example? 6
- C) What are the different types of wiring methods in electrification? 6
- Q5. Solve Any Two of the following.
- A) Describe working of air conditioning system with sketch ? 6
- B) Explain natural ventilation with sketch? 6
- C) What are the methods of thermal insulation used in building? 6
- Q6. Solve Any Two of the following.
- A) Explain various bye laws used in firefighting system of building? 6
- B) Define acoustics and its importance in building? 6
- C) Describe reverberation in acoustics? 6

*** End ***